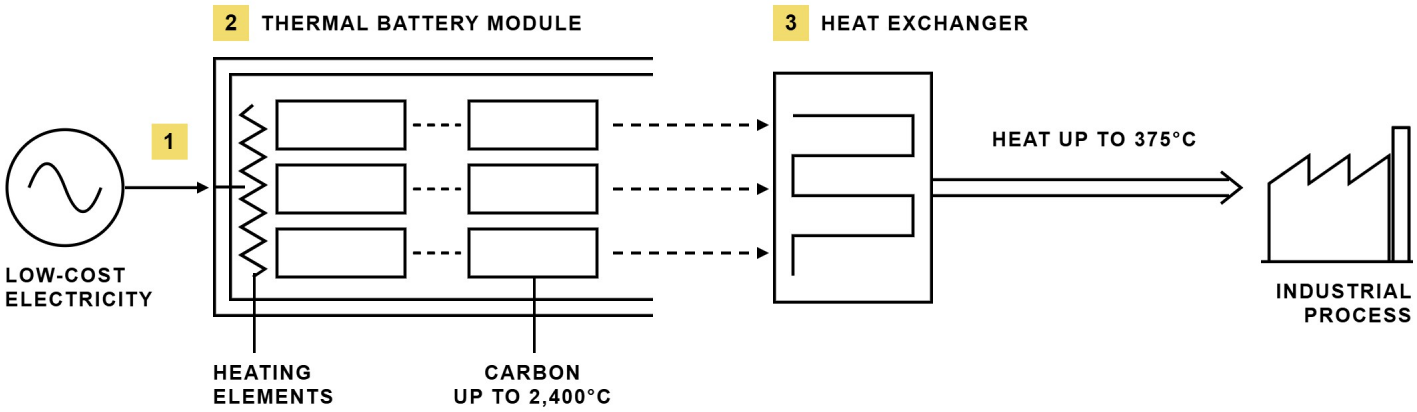
HOW ANTORA HEAT AND POWER WORK

Factory-built thermal batteries store low-cost electricity as heat in solid carbon and then deliver it as industrial heat or firm power.

HEAT — HEATCORE

UP TO 375°C · 300 kWth · CHARGES UP TO 900 kWe PER MODULE
ELECTRICITY → HEAT: UP TO 100% EFFICIENCY*



1 RAPID CHARGING

Each module resistively heats its carbon blocks, charging at up to 900 kWe.

2 MULTI-DAY STORAGE

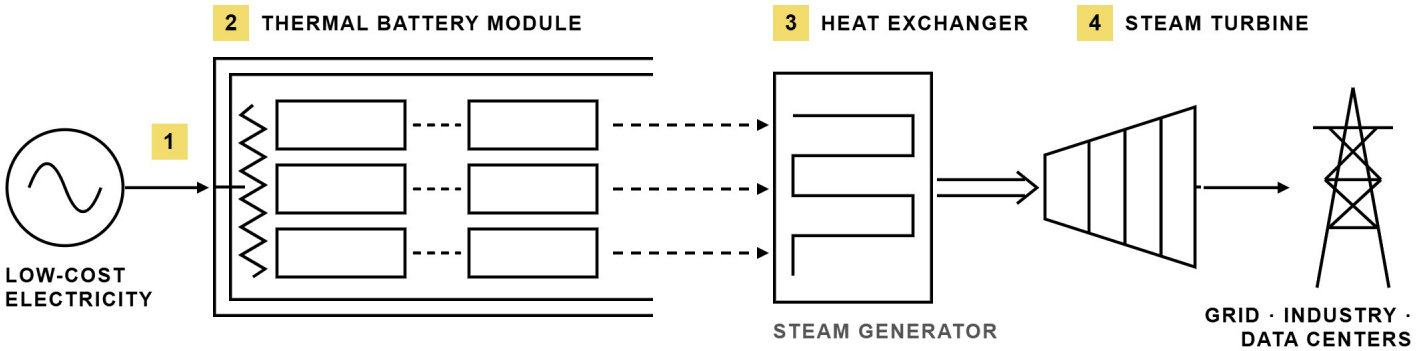
Carbon holds energy at up to 2,400°C; heat is delivered at up to 375°C.

3 HEAT ON DEMAND

Hot carbon radiates its energy as light, delivering process heat around the clock.

POWER — POWER BLOCK

50 MWe NAMEPLATE · CHARGES AT 50–500 MWe
HEAT → ELECTRICITY: ABOUT 33% EFFICIENCY*



1 RAPID CHARGING

Electricity heats the carbon blocks at 50–500 MWe, depending on the application.

2 MULTI-DAY STORAGE

Carbon blocks hold the energy for days, ready to dispatch as firm power.

3 STEAM

An attached steam generator turns stored heat into steam on demand.

4 GENERATE

A steam turbine delivers firm power: 50 MWe nameplate per power block.

NEXT Thermophotovoltaic (TPV) cells will convert light from the hot carbon to electricity at over 40% efficiency*, with no moving parts.

*Industry reference figures, not Antora specs: resistive heating up to 100% (Automation World); steam turbines about 33% (EPA CHP Catalog). TPV: Antora reports over 40%; MIT/NREL lab cells 41.1% at 2,400°C (Nature, 2022). Antora: antora.com, Oct 2026.